

# TI fields - subjects

## Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

## Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/EFmodel")
data <- read.csv("EFsubjects.csv")

# Assign factors

data$ID <- factor(data$ID)
data$ROI <- factor(data$ROI)
data$Region <- factor(data$Region)
data$StimType <- factor(data$StimType)
```

## EF - hippocampus & cortex - Table S2.1

```
# Calculating the average values for hippocampal and cortical regions; weighted by volume

# Hip TI 1:1
Hipp_TI <- subset(data, Region=="Hippocampus" & StimType=="TI 1:1")
Env_mod_amp_Hipp.11 <- normalise_by_total("Median", "ROI", "VolBrain", Hipp_TI)
sd_Env_mod_amp_Hipp.11 <- normalise_by_total("Std", "ROI", "VolBrain", Hipp_TI)
Abs_amp_Hipp.11 <- normalise_by_total("MedianTEf", "ROI", "VolBrain", Hipp_TI)
sd_Abs_amp_Hipp.11 <- normalise_by_total("StdTEf", "ROI", "VolBrain", Hipp_TI)

# Hip TI 1:3
Hipp_TI <- subset(data, Region=="Hippocampus" & StimType=="TI 1:3")
Env_mod_amp_Hipp.13 <- normalise_by_total("Median", "ROI", "VolBrain", Hipp_TI)
sd_Env_mod_amp_Hipp.13 <- normalise_by_total("Std", "ROI", "VolBrain", Hipp_TI)
Abs_amp_Hipp.13 <- normalise_by_total("MedianTEf", "ROI", "VolBrain", Hipp_TI)
sd_Abs_amp_Hipp.13 <- normalise_by_total("StdTEf", "ROI", "VolBrain", Hipp_TI)

# Cortex TI 1:1
```

```

Crtx_TI <- subset(data, Region=="Electrode" & StimType=="TI 1:1")
Env_mod_amp_Crtx.11 <- normalise_by_total("Median","ROI","VolBrain",Crtx_TI)
sd_Env_mod_amp_Crtx.11 <- normalise_by_total("Std","ROI","VolBrain",Crtx_TI)
Abs_amp_Crtx.11 <- normalise_by_total("MedianTEf","ROI","VolBrain",Crtx_TI)
sd_Abs_amp_Crtx.11 <- normalise_by_total("StdTEf","ROI","VolBrain",Crtx_TI)

# Cortex TI 1:3
Crtx_TI <- subset(data, Region=="Electrode" & StimType=="TI 1:3")
Env_mod_amp_Crtx.13 <- normalise_by_total("Median","ROI","VolBrain",Crtx_TI)
sd_Env_mod_amp_Crtx.13 <- normalise_by_total("Std","ROI","VolBrain",Crtx_TI)
Abs_amp_Crtx.13 <- normalise_by_total("MedianTEf","ROI","VolBrain",Crtx_TI)
sd_Abs_amp_Crtx.13 <- normalise_by_total("StdTEf","ROI","VolBrain",Crtx_TI)

Region <- c("Cortex","Cortex","Cortex","Cortex","Hippocampus","Hippocampus","Hippocampus","Hippocampus")
Stim <- c("TI 1:1", "TI 1:1", "TI 1:3", "TI 1:3", "TI 1:1", "TI 1:1", "TI 1:3", "TI 1:3")
Measure <- c("Env Mod Amp", "Abs Amp", "Env Mod Amp", "Abs Amp", "Env Mod Amp", "Abs Amp", "Env Mod Amp", "Abs Amp")
Median <- c(Env_mod_amp_Crtx.11, Abs_amp_Crtx.11, Env_mod_amp_Crtx.13, Abs_amp_Crtx.13, Env_mod_amp_Crtx.11, Abs_amp_Crtx.11, Env_mod_amp_Crtx.13, Abs_amp_Crtx.13)
SD <- c(sd_Env_mod_amp_Crtx.11, sd_Abs_amp_Crtx.11, sd_Env_mod_amp_Crtx.13, sd_Abs_amp_Crtx.13, sd_Env_mod_amp_Crtx.11, sd_Abs_amp_Crtx.11, sd_Env_mod_amp_Crtx.13, sd_Abs_amp_Crtx.13)

S2.1_Tot2mA <- data.frame(Stim,Measure,Region,Median,SD)
pander(S2.1_Tot2mA)

```

| Stim   | Measure     | Region      | Median  | SD      |
|--------|-------------|-------------|---------|---------|
| TI 1:1 | Env Mod Amp | Cortex      | 0.1167  | 0.08962 |
| TI 1:1 | Abs Amp     | Cortex      | 0.2305  | 0.1264  |
| TI 1:3 | Env Mod Amp | Cortex      | 0.09729 | 0.07292 |
| TI 1:3 | Abs Amp     | Cortex      | 0.2226  | 0.1278  |
| TI 1:1 | Env Mod Amp | Hippocampus | 0.1994  | 0.02476 |
| TI 1:1 | Abs Amp     | Hippocampus | 0.2366  | 0.02612 |
| TI 1:3 | Env Mod Amp | Hippocampus | 0.121   | 0.01563 |
| TI 1:3 | Abs Amp     | Hippocampus | 0.2333  | 0.02572 |

```

#
S2.1_Tot4mA <- data.frame(Stim,Measure,Region,Median*2,SD*2)
pander(S2.1_Tot4mA)

```

| Stim   | Measure     | Region      | Median...2 | SD...2  |
|--------|-------------|-------------|------------|---------|
| TI 1:1 | Env Mod Amp | Cortex      | 0.2334     | 0.1792  |
| TI 1:1 | Abs Amp     | Cortex      | 0.461      | 0.2529  |
| TI 1:3 | Env Mod Amp | Cortex      | 0.1946     | 0.1458  |
| TI 1:3 | Abs Amp     | Cortex      | 0.4451     | 0.2556  |
| TI 1:1 | Env Mod Amp | Hippocampus | 0.3987     | 0.04953 |
| TI 1:1 | Abs Amp     | Hippocampus | 0.4733     | 0.05223 |
| TI 1:3 | Env Mod Amp | Hippocampus | 0.2421     | 0.03126 |
| TI 1:3 | Abs Amp     | Hippocampus | 0.4667     | 0.05143 |

## EF - hippocampus - Figure 2C & Table S2.2

```
Hipp <- subset(data, Region=="Hippocampus")

## Normalise
Ratio_Total <- ddply(Hipp, c("ID","Ratio"), summarise,
  Ant_T = Median[ROI=="Ant"] / (Median[ROI=="Ant"] + Median[ROI=="Mid"] + Median[ROI=="Post"]),
  Mid_T = Median[ROI=="Mid"] / (Median[ROI=="Ant"] + Median[ROI=="Mid"] + Median[ROI=="Post"]),
  Post_T = Median[ROI=="Post"] / (Median[ROI=="Ant"] + Median[ROI=="Mid"] + Median[ROI=="Post"])
)

Ratio_long <- tidyr::gather(Ratio_Total, Regions, metric, Ant_T:Post_T, factor_key=TRUE)
Ratio_long$Regions <- as.factor(Ratio_long$Regions)

# Add this metric to the hippocampus data subset to also use for correlations with BOLD
ROI <- Ratio_long$Regions
ROI <- gsub("_T", "", ROI)
Ratio_long$ROI=ROI
rm(ROI)
Hipp <- merge(Hipp, Ratio_long,by=c("ID","ROI","Ratio"))

Hipp <- Hipp %>% rename(median_Region2Total = metric)

## EF subjects per hippocampal region
EF.values <- Hipp %>%
  group_by(StimType,ROI) %>%
  summarise_at(vars(median_Region2Total), list(median = mean, std = sd))

pander(EF.values)
```

| StimType | ROI  | median | std      |
|----------|------|--------|----------|
| TI 1:1   | Ant  | 0.3208 | 0.02938  |
| TI 1:1   | Mid  | 0.3696 | 0.01592  |
| TI 1:1   | Post | 0.3096 | 0.02752  |
| TI 1:3   | Ant  | 0.4041 | 0.02024  |
| TI 1:3   | Mid  | 0.3393 | 0.009827 |
| TI 1:3   | Post | 0.2565 | 0.01498  |

### ### Stats

#### # --- TI 1:1 - Table S2.2

```
TI1_1 <- subset(Hipp, StimType=="TI 1:1")
TI1_1.EF <- lmer(median_Region2Total ~ ROI + (1|ID), data = TI1_1, REML = TRUE)
Anova(TI1_1.EF, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: median_Region2Total
```

```
##           F Df Df.res   Pr(>F)
## ROI 26.045  2      30 2.77e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

TI1_1.EF.contrasts <- emmeans(TI1_1.EF, specs = pairwise ~ ROI, type = "response")
TI1_1.EF.contrasts

## $emmeans
## ROI emmean      SE df lower.CL upper.CL
## Ant  0.321 0.00625 45   0.308   0.333
## Mid  0.370 0.00625 45   0.357   0.382
## Post 0.310 0.00625 45   0.297   0.322
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Ant - Mid  -0.0487 0.00884 30  -5.515 <.0001
## Ant - Post  0.0113 0.00884 30   1.274 0.4203
## Mid - Post  0.0600 0.00884 30   6.789 <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates

# --- TI 1:3 - Table S2.2

TI1_3 <- subset(Hipp, StimType=="TI 1:3")
TI1_3.EF <- lmer(median_Region2Total ~ ROI + (1|ID), data = TI1_3, REML = TRUE)
Anova(TI1_3.EF, test="F")

## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: median_Region2Total
##           F Df Df.res   Pr(>F)
## ROI 359.62  2      30 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

TI1_3.EF.contrasts <- emmeans(TI1_3.EF, specs = pairwise ~ ROI, type = "response")
TI1_3.EF.contrasts

## $emmeans
## ROI emmean      SE df lower.CL upper.CL
## Ant  0.404 0.0039 45   0.396   0.412
## Mid  0.339 0.0039 45   0.331   0.347
## Post 0.257 0.0039 45   0.249   0.264
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
```

```

## $contrasts
## contrast estimate SE df t.ratio p.value
## Ant - Mid 0.0648 0.00552 30 11.741 <.0001
## Ant - Post 0.1476 0.00552 30 26.752 <.0001
## Mid - Post 0.0828 0.00552 30 15.011 <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates

### ---- Plot ----

p.ratio.TI <- ggplot(Hipp, aes(x = ROI, y = median_Region2Total, fill = ROI)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.6,
  scale_y_continuous(
    breaks = seq(0, .5, .1),
    limits=c(0, .5)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold", hjust = 1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Ant","Mid", "Post"),
    labels=c("Ant","Mid", "Post")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("") + scale_fill_manual(values=c("Ant","Mid","Post"))

ggsave(p.ratio.TI ,file="Figure_2C.pdf", width = 10.5, height = 9, dpi = 300, units = "cm", device = "pdf")

## -----
## Calculate ratios for Absolute amplitude
## Normalise
Ratio_Total <- ddply(Hipp, c("ID","Ratio"), summarise,
  Ant_T = MedianTEf[ROI=="Ant"] / (MedianTEf[ROI=="Ant"] + MedianTEf[ROI=="Mid"] + MedianTEf[ROI=="Post"]),
  Mid_T = MedianTEf[ROI=="Mid"] / (MedianTEf[ROI=="Ant"] + MedianTEf[ROI=="Mid"] + MedianTEf[ROI=="Post"]),
  Post_T = MedianTEf[ROI=="Post"] / (MedianTEf[ROI=="Ant"] + MedianTEf[ROI=="Mid"] + MedianTEf[ROI=="Post"])
)

Ratio_long <- tidyr::gather(Ratio_Total, Regions, metric, Ant_T:Post_T, factor_key=TRUE)
Ratio_long$Regions <- as.factor(Ratio_long$Regions)

# Add this metric to the hippocampus data subset to also use for correlations with BOLD
ROI <- Ratio_long$Regions
ROI <- gsub("_T", "", ROI)
Ratio_long$ROI=ROI
rm(ROI)

```

```
Hipp <- merge(Hipp, Ratio_long, by=c("ID", "ROI", "Ratio", "Regions"))

Hipp <- Hipp %>% rename(medianTEf_Region2Total = metric)
```

## BOLD - Import data and Organize it

```
# Here we have the %BOLD change for the contrast TI 1:1 > Sham and TI 1:3 > Sham
# only for the participants with EF models
BOLDdata <- read.csv("BOLD_Hipp_model11.csv")
```

```
# Assign factors
BOLDdata$ID <- factor(BOLDdata$ID)
BOLDdata$ROI <- factor(BOLDdata$ROI)
BOLDdata$hemisphere <- factor(BOLDdata$hemisphere)
BOLDdata$StimType <- factor(BOLDdata$StimType)
BOLDdata$TaskStage <- factor(BOLDdata$TaskStage)
```

```
# Match contrast names from model
BOLDdata$Stim <- NA
BOLDdata$Stim[BOLDdata$StimType=="TI>Sham"] <- "TI 1:1"
BOLDdata$Stim[BOLDdata$StimType=="T1st>Sham"] <- "TI 1:3"
```

```
BOLDdata <- subset(BOLDdata, select = -c(StimType))
```

```
BOLDdata <- BOLDdata %>% rename(
  StimType = Stim
)
```

```
BOLDdata$StimType <- as.factor(BOLDdata$StimType)
```

```
## Add ratio Region to total (normalisation)
```

```
Ratio_Total <- dply(BOLDdata, c("ID", "StimType"), summarise,
  Ant_T = BOLD[ROI=="Ant"] / (BOLD[ROI=="Ant"] + BOLD[ROI=="Mid"] + BOLD[ROI=="Post"]),
  Mid_T = BOLD[ROI=="Mid"] / (BOLD[ROI=="Ant"] + BOLD[ROI=="Mid"] + BOLD[ROI=="Post"]),
  Post_T = BOLD[ROI=="Post"] / (BOLD[ROI=="Ant"] + BOLD[ROI=="Mid"] + BOLD[ROI=="Post"])
```

```
Ratio_long <- tidyr::gather(Ratio_Total, Regions, metric, Ant_T:Post_T, factor_key=TRUE)
Ratio_long$Regions <- as.factor(Ratio_long$Regions)
```

```
# Add this metric to the hippocampus data subset to also use for correlations with BOLD
```

```
ROI <- Ratio_long$Regions
ROI <- gsub("_T", "", ROI)
Ratio_long$ROI=ROI
rm(ROI)
```

```
Hipp <- merge(Hipp, Ratio_long, by=c("ID", "ROI", "StimType", "Regions"))
```

```
Hipp <- Hipp %>% rename(BOLD_Region2Total = metric)
```

## Calculate Correlations between BOLD and TI EF

```
library(data.table)
```

```
## -- Hippocampal data ---
```

```
# Correlations - are the changes in BOLD in TI 1:3 related to the fields?
```

```
# median_Region2Total = Normalised envelope modulation amplitude
```

```
TI13 <- subset(Hipp, StimType=="TI 1:3")
```

```
ddply(TI13, .(ROI), corrgroup, "BOLD_Region2Total", "median_Region2Total")
```

```
##      ROI      COR      p.value
## 1  Ant -0.61828668 0.010683368
## 2  Mid -0.07330268 0.787322958
## 3 Post -0.63452802 0.008283665
```

```
# Yes
```

```
# Could this also be explained by the absolute amplitude of the field?
```

```
# medianTEf_Region2Total = Normalised absolute amplitude
```

```
ddply(TI13, .(ROI), corrgroup, "BOLD_Region2Total", "medianTEf_Region2Total")
```

```
##      ROI      COR      p.value
## 1  Ant -0.28184214 0.2902606
## 2  Mid -0.07588448 0.7800008
## 3 Post -0.45699921 0.0751392
```

```
# No
```

```
## Also checking if correlations exist for TI 1:1
```

```
TI11 <- subset(Hipp, StimType=="TI 1:1")
```

```
ddply(TI11, .(ROI), corrgroup, "BOLD_Region2Total", "median_Region2Total")
```

```
##      ROI      COR      p.value
## 1  Ant -0.1062482 0.69533172
## 2  Mid  0.1807378 0.50294867
## 3 Post -0.4917825 0.05301215
```

```
ddply(TI11, .(ROI), corrgroup, "BOLD_Region2Total", "medianTEf_Region2Total")
```

```
##      ROI      COR      p.value
## 1  Ant  0.16840795 0.5329777
## 2  Mid -0.05352784 0.8439190
## 3 Post -0.21951387 0.4140057
```

```
## -- Could the changes in BOLD in the hippocampus during TI 1:3 be explained by the field's absolute a
```

```
CorrAntHipp <- subset(Hipp, ROI=="Ant" & StimType=="TI 1:3")
```

```
CorrAntHipp <- CorrAntHipp %>% dplyr::select(ID, BOLD_Region2Total, median_Region2Total, medianTEf_Regi
```

```

CortexTEf <- subset(data, Region=="Electrode" & StimType=="TI 1:3")
Cortex.Ant <- CortexTEf$MedianTEf[CortexTEf$ROI=="Ant"]
Cortex.Mid <- CortexTEf$MedianTEf[CortexTEf$ROI=="Mid"]
Cortex.Post <- CortexTEf$MedianTEf[CortexTEf$ROI=="Post"]

CorrAntHipp$Cortex.Ant <- Cortex.Ant
CorrAntHipp$Cortex.Mid <- Cortex.Mid
CorrAntHipp$Cortex.Post <- Cortex.Post
corr.test(CorrAntHipp$BOLD_Region2Total, CorrAntHipp$Cortex.Ant)

## Call:corr.test(x = CorrAntHipp$BOLD_Region2Total, y = CorrAntHipp$Cortex.Ant)
## Correlation matrix
## [1] -0.01
## Sample Size
## [1] 16
## These are the unadjusted probability values.
## The probability values adjusted for multiple tests are in the p.adj object.
## [1] 0.97
##
## To see confidence intervals of the correlations, print with the short=FALSE option

corr.test(CorrAntHipp$BOLD_Region2Total, CorrAntHipp$Cortex.Mid)

## Call:corr.test(x = CorrAntHipp$BOLD_Region2Total, y = CorrAntHipp$Cortex.Mid)
## Correlation matrix
## [1] -0.02
## Sample Size
## [1] 16
## These are the unadjusted probability values.
## The probability values adjusted for multiple tests are in the p.adj object.
## [1] 0.94
##
## To see confidence intervals of the correlations, print with the short=FALSE option

corr.test(CorrAntHipp$BOLD_Region2Total, CorrAntHipp$Cortex.Post)

## Call:corr.test(x = CorrAntHipp$BOLD_Region2Total, y = CorrAntHipp$Cortex.Post)
## Correlation matrix
## [1] -0.14
## Sample Size
## [1] 16
## These are the unadjusted probability values.
## The probability values adjusted for multiple tests are in the p.adj object.
## [1] 0.6
##
## To see confidence intervals of the correlations, print with the short=FALSE option

```

## Plot Correlations between BOLD and TI EF - Figure S7

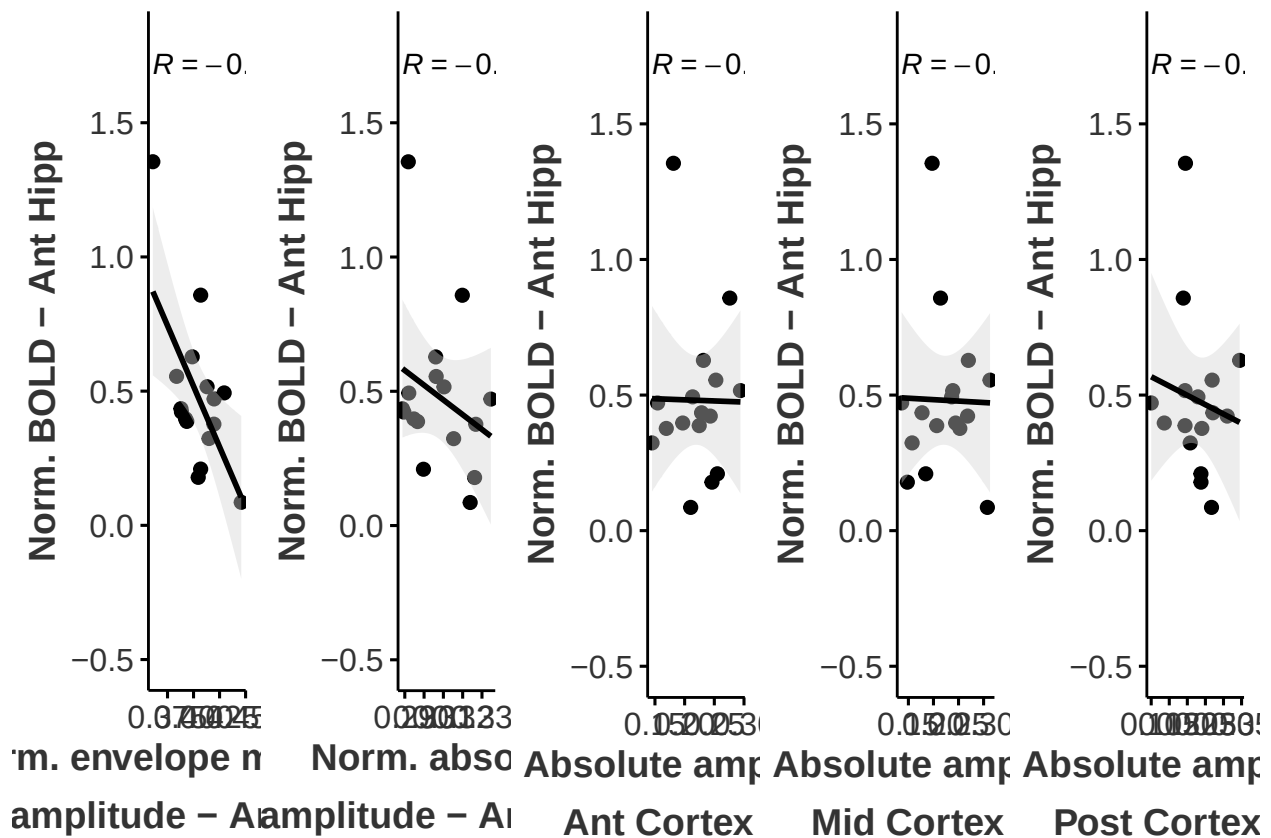
```

# Stim Type = TI 1:3
# -- Ant Hippocampus BOLD
vars = c("median_Region2Total", "medianTEf_Region2Total", "Cortex.Ant", "Cortex.Mid", "Cortex.Post")
xname = c(expr(bold(atop("Norm. envelope modulation", paste("amplitude - Ant Hipp")))), expr(bold(atop(
resp = tibble(vars, xname)

library(purrr)
allplots = pmap(resp, ~scatter_fun2(x = .x, y = "BOLD_Region2Total", ROIy = "Ant", xname = .y, -0.5, 1.5))

all_corr_AntBOLD <- grid.arrange(allplots[[1]], allplots[[2]], allplots[[3]], allplots[[4]], allplots[[5]])

```



Plot Correlations between BOLD and TI EF - Figure S7

```

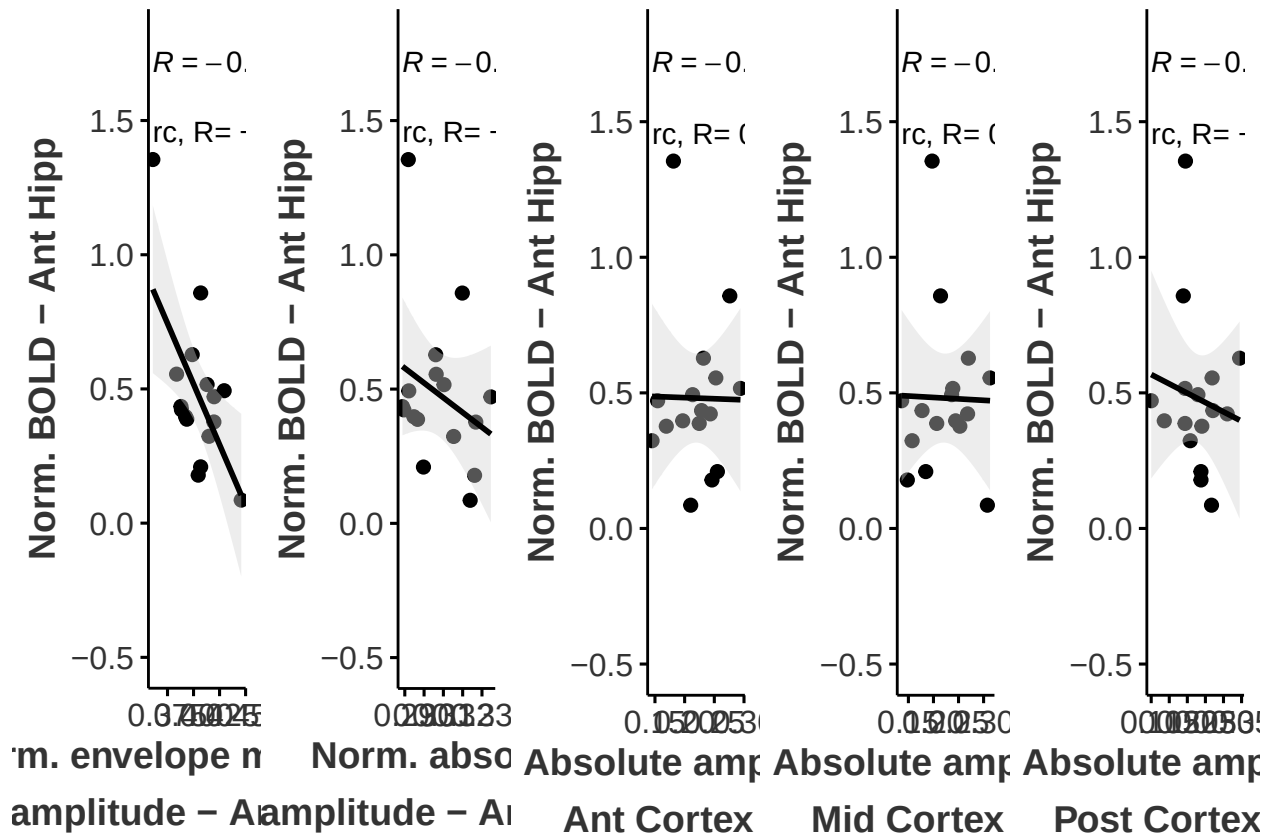
# Stim Type = TI 1:3
# -- Ant Hippocampus BOLD

vars = c("median_Region2Total", "medianTEf_Region2Total", "Cortex.Ant", "Cortex.Mid", "Cortex.Post")
xname = c(expr(bold(atop("Norm. envelope modulation", paste("amplitude - Ant Hipp")))), expr(bold(atop(
resp = tibble(vars, xname)

library(WRS2)
allplots = pmap(resp, ~scatter_fun3(x = .x, y = "BOLD_Region2Total", ROIy = "Ant", xname = .y, -0.5, 1.5))

```

```
all_corr_AntBOLD <- grid.arrange(allplots[[1]], allplots[[2]], allplots[[3]], allplots[[4]], allplots[[5]])
```



```
ggsave(all_corr_AntBOLD, file="Figure_S7A.pdf", width = 45, height = 8, dpi = 300, units = "cm", device = "pdf")

# -- Mid Hippocampus BOLD

CorrMidHipp <- subset(Hipp, ROI=="Mid" & StimType=="TI 1:3")
CorrMidHipp <- CorrMidHipp %>% dplyr::select(ID, BOLD_Region2Total, median_Region2Total, medianTEf_Region2Total)

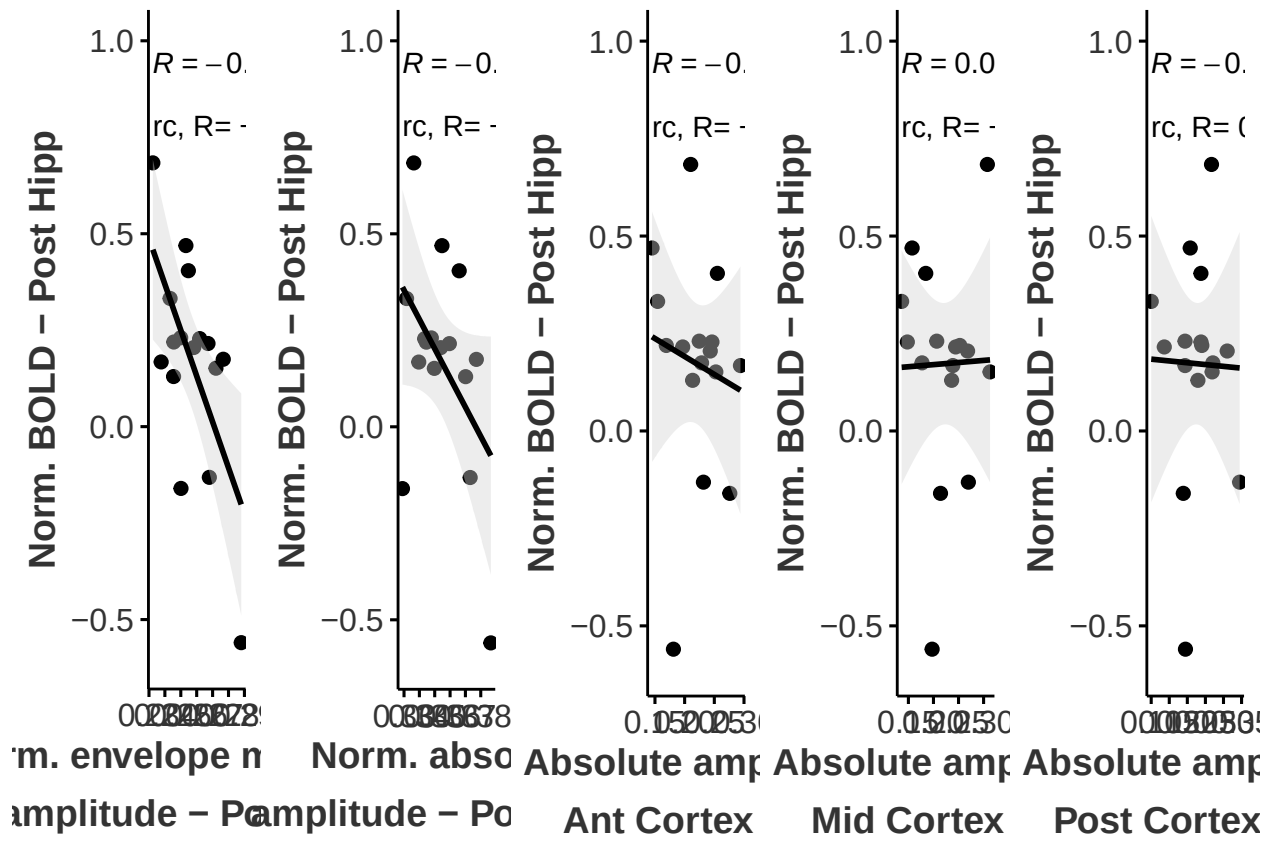
CorrMidHipp$Cortex.Ant <- Cortex.Ant
CorrMidHipp$Cortex.Mid <- Cortex.Mid
CorrMidHipp$Cortex.Post <- Cortex.Post

vars = c("median_Region2Total", "medianTEf_Region2Total", "Cortex.Ant", "Cortex.Mid", "Cortex.Post")
xname = c(expr(bold(atop("Norm. envelope modulation", paste("amplitude - Mid Hipp")))), expr(bold(atop("amplitude - Ant Cortex", paste("amplitude - Ant Cortex")))), expr(bold(atop("amplitude - Mid Cortex", paste("amplitude - Mid Cortex")))), expr(bold(atop("amplitude - Post Cortex", paste("amplitude - Post Cortex")))))
resp = tibble(vars, xname)

allplots = pmap(resp, ~scatter_fun3(x = .x, y = "BOLD_Region2Total", ROIy = "Mid", xname = .y, 0, 0.8, 0.05))

all_corr_MidBOLD <- grid.arrange(allplots[[1]], allplots[[2]], allplots[[3]], allplots[[4]], allplots[[5]])
```





```
ggsave(all_corr_PostBOLD ,file="Figure_S7C.pdf", width = 45, height = 8, dpi = 300, units = "cm", device = "pdf")
```